Effects of Pre-race Apneas on 400-m Freestyle Swimming Performance

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Abstract

Robertson, C, Lodin-Sundström, A, O'Hara, J, King, R, Wainwright, B, and Barlow, M. Effects of pre-race apneas on 400-m freestyle swimming performance. *J Strength Cond Res* 34(3): 828–837, 2020—This study aimed to establish whether a series of 3 apneas before a 400-m freestyle time-trial affected swimming performance when compared with and combined with a warm-up. Nine (6 males and 3 females) regional to national standard swimmers completed four 400-m freestyle time-trials in 4 randomized conditions: without warm-up or apneas (CON), warm-up only (WU), apneas only (AP), and warm-up and apneas (WUAP). Time-trial performance was significantly improved after WUAP (275.79 ± 12.88 seconds) compared with CON (278.66 ± 13.31 seconds, $p = 0.035$) and AP (278.64 ± 4.10 seconds, $p = 0.015$). However, there were no significant differences between the WU (276.01 ± 13.52 seconds, $p > 0.05$) and other interventions. Spleen volume compared with baseline was significantly reduced after the apneas by a maximum of ~45% in the WUAP and by ~20% in WU. This study showed that the combination of a warm-up with apneas could significantly improve 400-m freestyle swim performance compared with a control and apnea intervention. Further investigation into whether long-term apnea training can enhance this response is justified.

Key Words breath holding, hypoxia, splenic contraction, warm-up

Introduction

The difference between a Gold and Silver medal in the men's 400-m freestyle at the 2016 Rio Summer Olympic Games was 0.13 second. This emphasizes the point that in elite swimming, marginal improvements are a key priority (10). At Olympic and World Championship level, many swimmers have similar performance abilities determined by combinations of energy availability and technical effectiveness. Therefore, finding improvements in the performance of as little as 1% will have a significant effect on the race outcome (15).

The use of a warm-up is now widely accepted as common practice before an exercise bout (22). Two functions of a warm-up are to decrease the risk of injury and to prepare the athlete optimally for the demands of the event. This occurs by increasing muscle and body temperature, decreasing viscous resistance, increasing the rate of nerve impulses, and increasing efficiency of oxygen delivery (22). The effect of body temperature on exercise performance has recently been shown (16,24). The researchers found a significant improvement in jump height (3.8%) and 10-m sprint time (5.5%) when body temperature was increased after a warm-up. Further research has found that a specific warm-up reduced viscous resistance significantly by a shortening of the time taken for voluntary contractions (8).

Research into the effects of an active warm-up on swimming performance has sometimes been contradictory; however, it is suggested that the warm-up becomes more effective as the distance of the time-trial increases (22).

Apnea has been proposed as a new potential training method that would influence performance (18). Through holding one's breath, a number of physiological processes occur. Because of the reduction of oxygen availability during an apnea, hypoxemia occurs in the kidney and spleen. This response occurs in conjunction with hypercapnia, increased acidosis, bradycardia, reduced SpO_2, and a splenic contraction (4). During an initial apnea, the hypoxemia is one of the potential stimuli for a splenic contraction. This contraction usually occurs as a stress response, especially when the demands of oxygen increase, and is correlated with the release of catecholamines from the adrenal gland and the postganglionic fibers in the sympathetic nervous system (23).

There is an improvement in the body's ability to transport oxygen because of an increased number of circulating erythrocytes, resultant of a splenic contraction. This has been confirmed when a 2–5% increase in hemoglobin (Hb) and hematocrit (Hct), independent of hemoconcentration, and

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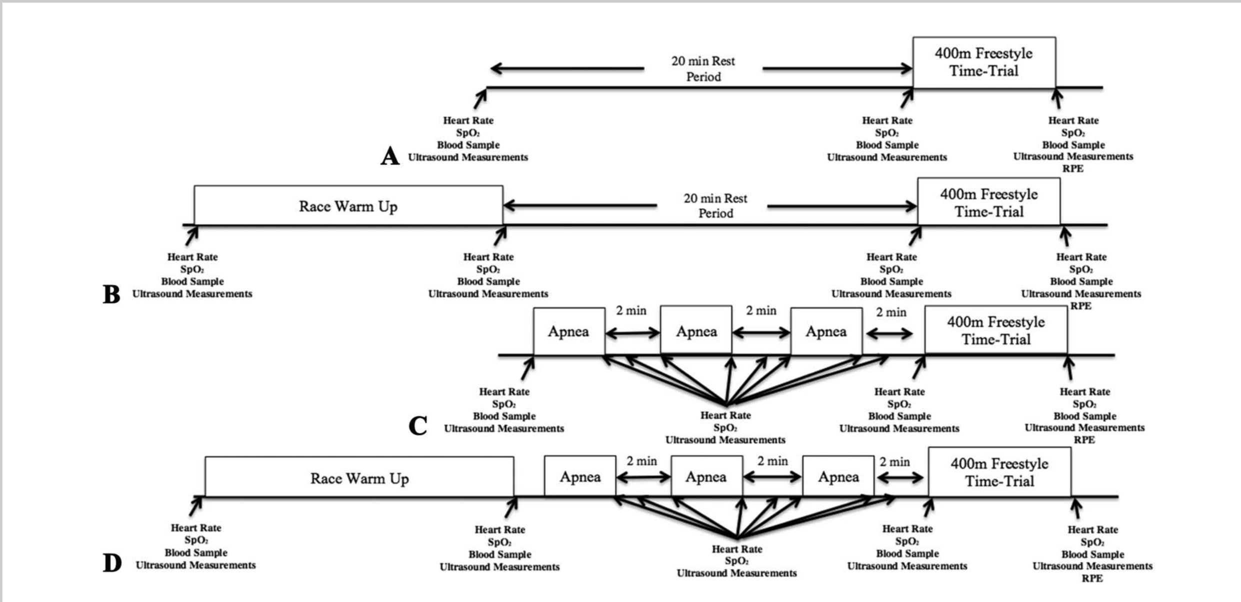

Figure 1. Protocols used in each intervention with measurement time points. Participants completed the interventions at least 2 days apart, and a maximum of 7 days. A) Control (no warm-up/apneas), (B) warm-up, (C) apneas, and (D) warm-up and apneas combined. RPE = rate of perceived exertion.

a reduction in arterial oxygen desaturation were seen after repeated apneas (3). This effect was confirmed when comparing splenectomized subjects against nonsplenectomized subjects. Subjects without a spleen did not demonstrate the same hematological changes after an apnea, or the ability to perform the typical prolongation of repeated apneas, compared with their nonsplenectomized counterparts (3,4).

The strength of the splenic contraction has been suggested to be augmented by a series of apneas rather than a single breath hold, although a study found that a splenic contraction could potentially occur after a single apnea (4). However, in the same study, further apneas continued to steadily reduce splenic volume, thereby further increasing the circulating erythrocytes. The increase in circulating erythrocytes has been shown to last between 8 and 9 minutes after serial apneas (27), suggesting a potential method of acutely increasing oxygen transport, to improve short-term competitive performance.

From the involvement of both the central inspiratory and phasic pulmonary afferent mechanisms during apneas, heart rate (HR) becomes slower (bradycardia). This is an essential oxygen conserving part of the diving response. The diving response can be augmented by cold-water (~10° C) face immersion after 25–30 seconds (19). This occurs due to the stimulation of the ophthalmic branch of the trigeminal nerve, situated in the upper facial region (2). This sensory output has therefore suggested to be beneficial for free divers (20), but also possibly swimmers who spend a large proportion of the race with their face immersed.

Based on the physiological research of the diving response and splenic contraction, a plausible assumption can be made that the acute effects of serial apneas can improve aerobic performance.

The study by Sperlich et al. (30) was the first and currently the only study to apply the potential acute effects of repeated apneas before a cycling time-trial. The results showed no improvement in performance, as well as no rise in Hct or Hb, contradicting previous findings.

It is clear that there are gaps in the research regarding the effects of a series of apneas on 400-m freestyle time-trial performance. Because of the potential acute physiological effects that the apneas may have on aerobic exercise, it seems logical for these to be placed immediately before the time-trial, adding to the potential effects already gained by the warm-up.

Table 1. Shows the standardized warm-up used before the 400-m freestyle time-trial in the warm-up intervention, and warm-up and apnea combined intervention.

400-m freestyle	30-s rest
200-m pull	30-s rest
200-m kick	30-s rest
200-m drill	30-s rest
200-m individual medley	30-s rest
4 × 50 at 400-m race pace	On 1 min
200-m steady	

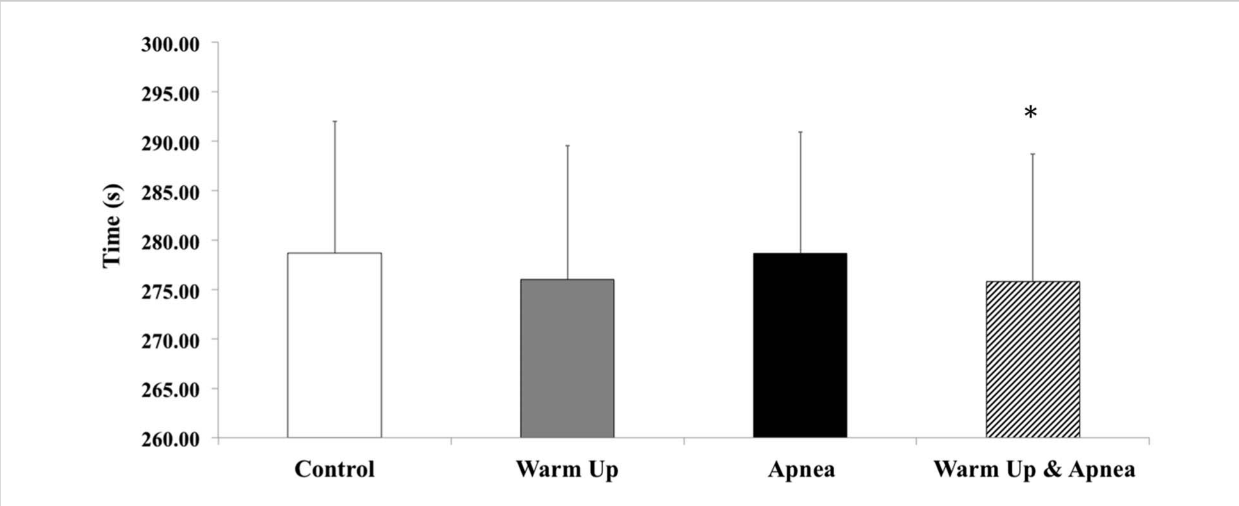

Figure 2. Time taken to complete the 400-m freestyle. Warm-up and apnea intervention were significantly faster than the control and apnea intervention. Error bars are represented as the *SD*. Significant differences are denoted by $^*p < 0.05$.

The aim of this research is to establish whether a series of apneas immediately before a 400-m freestyle time-trial affects performance when compared with and combined with a warm-up.

Methods

Experimental Approach to the Problem

A randomized, repeated-measure cross-over design was used for this study. The testing took place during an aerobic capacity period of training when the pool was configured for short-course (25 m). This setup included an indoor pool (water temperature, 28.37 ± 0.26° C; air temperature, 29.13 ± 0.42° C) consisting of antiwave lane ropes (AntiWave Pool Products, Constantine, MI, USA), OMEGA OSB11 starting blocks (Swiss Timing Ltd., Corgémont, Switzerland), and OMEGA OCP5 Touchpad's (Swiss Timing Ltd.). A familiarization session occurred before any testing to allow participants to habituate themselves with the apnea protocol (Figure 1).

Participants were subjected to 4 separate test conditions: no warm-up/no apneas (CON), with warm-up only (WU), apneas only (AP), and with warm-up and apneas (WUAP), on a 400-m freestyle time-trial. Visits were separated by at least 2 days and a maximum of 7 days.

Before each time-trial, the participants were asked to refrain from strenuous exercise (24 hours), caffeine (12 hours), and food (2 hours). Swim and land training sessions were controlled (low intensity, the rate of perceived exertion [RPE] <12 (21)) and replicated 2 days leading into the trials as to not affect the performance outcome. In addition, participants were asked to replicate their dietary intake for the 24 hours before each time-trial.

Subjects

Nine well-trained swimmers (6 males and 3 females: age 19 ± 1 and 19 ± 2 years; stature 1.84 ± 0.05 and 1.67 ± 0.03 m; and body mass 78.2 ± 7.2 and 61.6 ± 2.2 kg, respectively; all items measured as mean ± *SD*) volunteered to participate in this study. Personal best times in the 400-m freestyle event were 257.99 ± 5.58 and 262.06 ± 13.90 seconds for the male and female participants, respectively. Participants were aged between 16 and 22 years and had competed at a regional/national level in middle-distance swimming events. They were free from any illnesses and injuries that may have affected their ability to compete in this study. Informed consent was given by all participants. Written parental consent was given for any participants aged under 18 years of age in addition to their own written consent. This study was approved by the Leeds Beckett University Institutional Ethical Committee.

Procedures

The first visit was a familiarization session to habituate the participants to the study protocol, as none of the participants had any previous breath holding experience. This included performing the full apnea protocol (Figure 1) until they were confident with the procedure. The participants were screened when they first arrived to ensure that they were healthy enough to complete the task. This screening protocol included a questionnaire established by using the American College of Sports Medicine (ACSM) Guidelines (1), and measurements of the participants' stature, mass, and blood pressure were taken. Body mass was recorded to the nearest 0.1 kg using Seca Scales 709 (Seca Ltd., Birmingham, United Kingdom). Stature was measured using a Harpeden Portable Stadiometer (Holtain Limited, Pembrokeshire, United Kingdom) to the nearest 0.01 m.

During experimental trials, 4 different types of pre-time-trial condition were followed by an individual 400-m freestyle time-trial from a dive start. The order in which the participant performed these trials was randomized. The

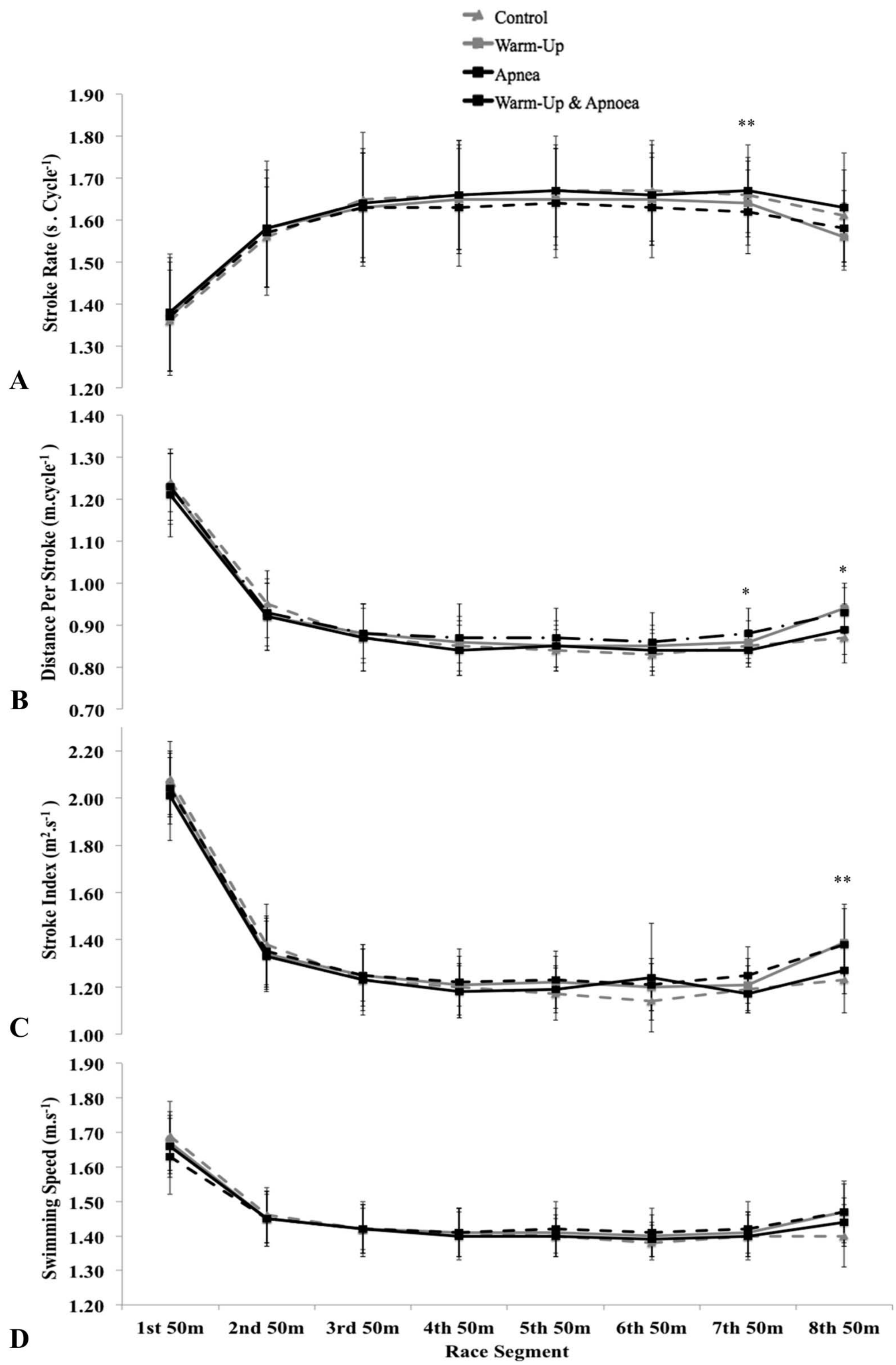

Figure 3. Shows the race analysis and stroke parameters for each 50-m of the 400-m freestyle. A) Shows the swimming speed ($m \cdot s^{-1}$). B) Shows the stroke rate ($s \cdot cycle^{-1}$). C) Shows the distance per stroke (DPS). D) Shows the stroke index ($m^2 \cdot s^{-1}$) worked out by multiplying the DPS by the swimming speed. Error bars represent the $\pm SD$. Significant difference is denoted by $*p < 0.05$ and $**p < 0.01$.

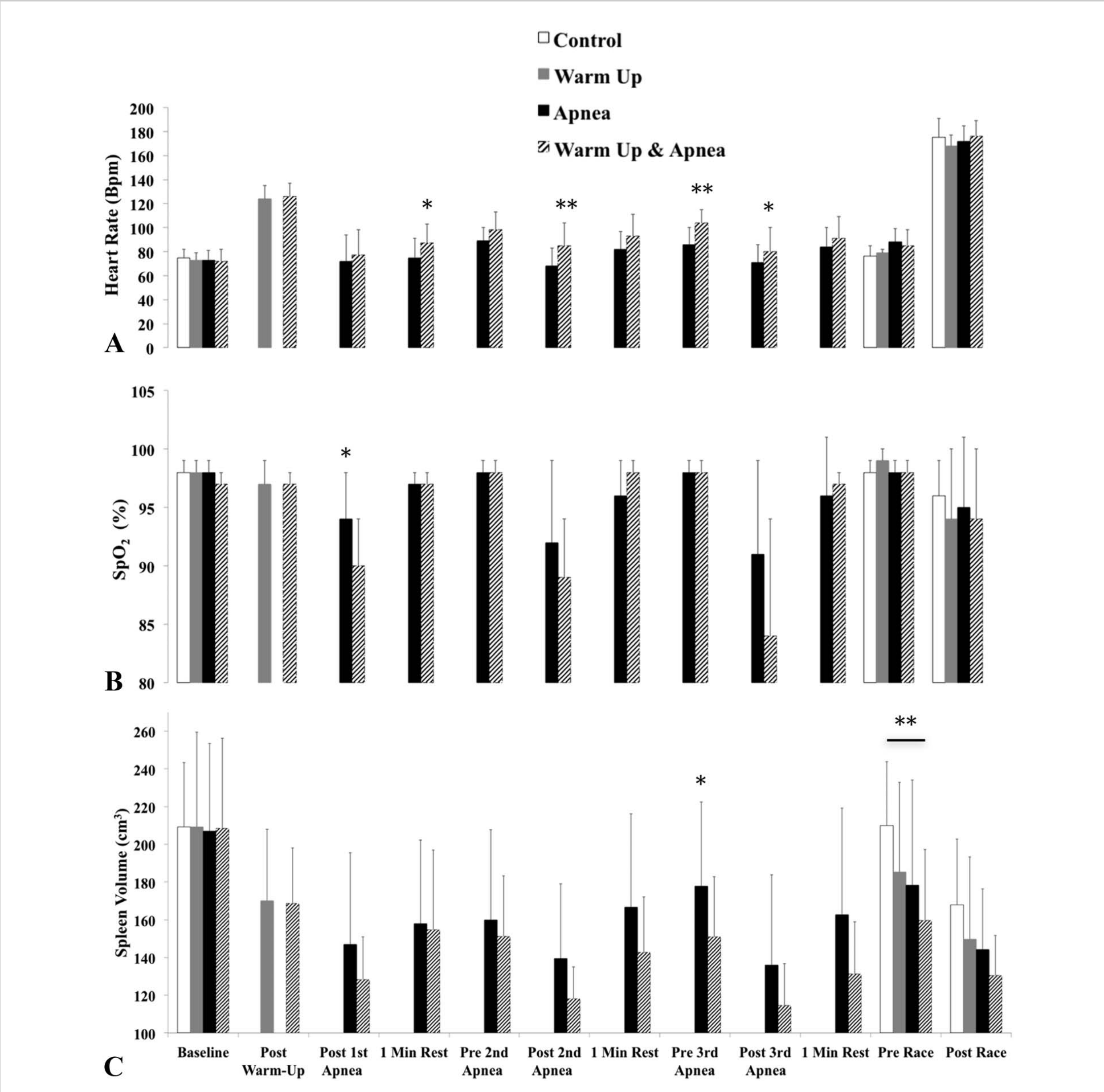

Figure 4. A) Heart rate, (B) SpO_2 (error bars are represented as the upper limits of the 95% CI because of the large size of the *SD*), and (C) spleen volume at different time points during the interventions. For (B and C), error bars represent the +*SD*. Significance is denoted by *$p < 0.05$ and **$p < 0.01$. Because of conditions A and B having no apneas, there are no data at these points. Because of conditions A and C not including a warm-up, there are no data at these points. CI = confidence interval.

participants performed their time-trial at the same time of day to ensure that their normal circadian rhythms did not affect the results. A schematic of the protocols can be seen in Figure 1.

Condition A was a control trial where neither the apnea intervention nor pre-race warm-up occurred. Physiological measurements of HR (Polar V800 HR monitor; Polar Electro, Kempele, Finland), SpO_2 (Nonin 8500 Hand Held Digital Oximeter, Plymouth, MA, USA), blood lactate (BL; Lancet and Lactate Plus Meter; Nova Biomedical Corporation, Waltham, MA, USA), Hb (HemoCue, Radiometer Group, Ängelholm, Sweden) taken from the ear, and spleen volume (Vivid I, GE Healthcare, General Electric, Fairfield, CT, USA) were all taken in a seated position at baseline, before, and immediately after time-trial with the addition of RPE (6) taken after time-trial in all conditions.

Condition B introduced the standardized race warm-up (Table 1). Here, the physiological measurements were taken

Table 2. Shows mean ± *SD* (95% CI) of blood lactate ($mmol \cdot L^{-1}$) in all 4 interventions at different time points during the testing protocol.

Intervention	Time point			
	Baseline	Post-warm-up	Pre-time-trial	Post-time-trial
Control	1.23 ± 0.33 (0.97–1.49)		1.18 ± 0.47 (0.82–1.55)	9.84 ± 3.67 (7.02–12.66)
Warm-up	1.27 ± 0.32 (1.02–1.52)	3.20 ± 1.35 (2.17–4.24)	1.67 ± 0.55 (1.25–2.09)	11.24 ± 2.54 (9.29–13.19)
Apnea	1.26 ± 0.61 (0.79–1.73)		1.16 ± 0.45 (0.81–1.50)	9.52 ± 2.12 (7.89–11.14)
Warm-up and apnea	1.01 ± 0.49 (0.64–1.39)	2.29 ± 0.88 (1.61–2.97)	1.43 ± 0.68 (0.90–1.96)	9.11 ± 2.59 (7.11–11.10)

before and after warm-up, and before and immediately after time-trial. In both conditions A and B, there was a 20-minute seated passive rest period to stimulate a call room before the time-trial. During this passive rest, participants remained in their racing suits plus a towel that covered their bodies from their shoulders to waist, to ensure that different items of clothing affected the results.

Condition C involved the apnea intervention independent of a warm-up. The apnea intervention consisted of 3 seated maximal breath-holds, until voluntary cessation. Between each apnea, there was a 2-minute passive rest period. Physiological measurements were taken at baseline, before, and immediately after time-trial. Heart rate, SpO_2, and spleen volume was measured before, after, and 1-minute after each apnea.

Condition D was a combination of the interventions where both a standardized race warm-up and a series of apneas were performed before the 400-m time-trial.

The 400-m freestyle time-trial splits were taken every 50 m by using the OMEGA OCP5 Touchpads. Time-trial analysis (stroke rate [SR], distance per stroke [DPS], and stroke index [SI]) was performed during and immediately after the time-trial. Stroke rate was obtained using the stopwatch function on the Finis 3 × 300 m stopwatch (Finis, Inc., CA, USA). This was computed by calculating the elapsed time for 3 stroke cycles. The timing started as the swimmers right hand entered the water and then stopped after the same hand had entered the water for the fourth time, thereby completing the 3 arm cycles. The measurements were taken between 10 and 20 m to eliminate the effects of the start and turns. The SR values were then divided by 60 to get the SR per cycle ($s \cdot cycle^{-1}$). Distance per stroke and SI were obtained using a method devised by Vitor and Böhme (32). The 50-m splits from the time-trial were converted to swimming speed ($m \cdot s^{-1}$), which allowed for DPS and SI to be calculated. Using the equation: swimming speed divided by SR, this gave a value for DPS ($m \cdot cycle^{-1}$). Stroke index ($m^2 \cdot s^{-1}$) was then calculated by multiplying swimming speed ($m \cdot s^{-1}$) by DPS.

Statistical Analyses

Statistical analysis was performed using SPSS software (version 22: SPSS, Inc., Chicago, IL, USA), with significance set at $p \leq 0.05$. Time-trial data were analyzed using a repeated-measures analysis of variance with a post hoc Bonferroni correction to compare differences between the

Table 3. Shows the mean ± *SD* (95% CI) hemoglobin ($g \cdot L^{-1}$) levels at different time points during the 4 interventions.

Intervention	Time point			
	Baseline	Post-warm-up	Pre-time-trial	Post-time-trial
Control	155 ± 10 (143–160)		149 ± 11 (141–157)	153 ± 12 (144–162)
Warm-up	150 ± 13 (140–161)	150 ± 11 (142–159)	151 ± 15 (140–163)	153 ± 10 (146–161)
Apnea	153 ± 12 (144–164)		153 ± 9 (146–163)	150 ± 13 (141–160)
Warm-up and apnea	151 ± 15 (139–164)	150 ± 10 (143–159)	150 ± 10 (143–157)	152 ± 16 (140–164)

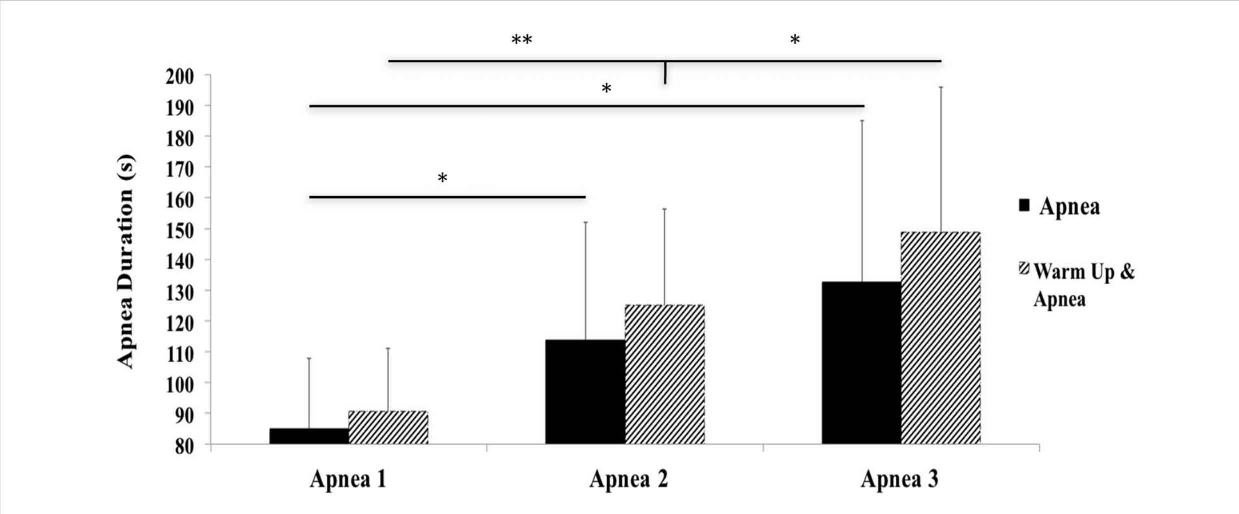

Figure 5. Shows the apnea duration in the apnea intervention, and warm-up and apnea combined intervention. Error bars represent the +*SD*. Significant differences are denoted by $^{*}p < 0.05$ and $^{**}p < 0.01$.

conditions. A pair-sample *t*-test was used when only 2 conditions were being compared.

Magnitude-based inferences (MBIs) were calculated to establish whether the effects of the interventions on the time-trial were meaningful (5). Hopkins (5) created a spreadsheet that was used by converting the *p* value into 90% confidence intervals (CIs). To establish whether the effect was beneficial, trivial, or harmful, verbal descriptors were used according to the following scale: <0.5%, "almost certainly not"; 0.5–5%, "very unlikely not"; 5–25%, "unlikely"; 25–75%, "possibly"; 75–95%, "likely"; 95–99.5%, "very likely"; and >99.5%, "almost certainly." When an odds ratio of benefit to harm of <66 was identified, then, the effect was deemed unclear; this corresponds to a 25% chance of benefit and 0.5% risk of harm (11). Based on the reduced inferential error rates compared with null-hypothesis significance testing, MBI has been supported within exercise science and is used to facilitate direct interpretation of the magnitude of changes and whether these are meaningful (13). Subsequently, this approach was used and prioritized for evaluating time-trial performance.

Twenty swimmers (10 males and 10 females) from the British swimming results database who have raced the 400-m freestyle in the past 12 months were analyzed to establish the typical error of this event. The swims were only included if it was in a short-course pool (25-m) and was in the aerobic training phase (high volume and low intensity–phase agreed with by coach). The analysis using Hopkins "Spreadsheets for Analysis of Validity and Reliability" (12) gave a log-transformed typical error as a coefficient of variation, which was 0.6% between trials. All data are presented as mean ± *SD*.

Results

Time-Trial Performance

The 400-m freestyle time-trial results are shown in Figure 2. The time swam in the WUAP time-trial (275.79 ± 12.88 seconds) was significantly faster than the CON (278.66 ± 13.31 seconds, $p = 0.035$) and the AP time-trial (278.64 ± 4.10 seconds, $p = 0.015$). With a mean difference between the CON and WUAP of 2.87 seconds, the intervention was deemed very likely beneficial (96.0%), very unlikely trivial (3.2%), and very unlikely negative (0.8%). There was no significant difference between the WU (276.01 ± 13.52 seconds) and any of the other conditions. Although it was not significant ($P = 0.097$) against the CON intervention, the WU intervention was still deemed likely beneficial when analyzed using MBI (likely beneficial [90.8%], unlikely trivial [6.7%], and very unlikely negative [2.5%]). When compared with CON, the experimental conditions resulted in the following performance improvements: 1.03% (WUAP), 0.95% (WU), and 0.01% (AP). Comparing the WUAP and WU conditions, there were no significant differences ($P = 0.99$) with the WUAP intervention being deemed unclear (possibly beneficial [50%] or possibly negative [50%]).

Stroke parameters (SR, DPS, and SI) and swimming speed are shown in Figure 3A–D. Stroke rate was significantly higher ($p = 0.002$) on the seventh 50 m, and DPS was significantly greater on the seventh ($p = 0.036$) and eighth 50 m ($P = 0.011$) for WUAP than in the AP intervention. Stroke index was found to be higher in the WUAP than the CON throughout the trial, significantly in the eighth 50 m ($p = 0.038$).

A significant difference was found between the post-trial RPE scores, with lower scores being reported in the WU time-trial (17.6 ± 0.53) compared against the CON time-trial (18.4 ± 0.53, $p = 0.013$) and AP time-trial (18.3 ± 0.50, $p = 0.004$). Yet, no difference was found between the WU and WUAP (17.9 ± 0.78, $p = 0.99$).

Physiological Results

Heart rate (Figure 4A) was significantly lower in the AP compared with the WUAP in the 1-minute rest after the first apnea (75 ± 16 vs. 87 ± 16 $\text{b}\cdot\text{min}^{-1}$; $p = 0.04$), the second apnea (68 ± 15 vs. 85 ± 19 $\text{b}\cdot\text{min}^{-1}$, $p = 0.003$), before the third apnea (86 ± 14 vs. 104 ± 11 $\text{b}\cdot\text{min}^{-1}$, $p = 0.002$), and finally, after the third apnea (71 ± 15 vs. 80 ± 20 $\text{b}\cdot\text{min}^{-1}$, $p = 0.047$). No other significant differences were found.

In the AP intervention, SpO_2 (Figure 4B) was significantly higher (94 ± 4%) than the WUAP (90 ± 4%) after the first apnea ($p = 0.03$). No other differences were found.

Spleen volume (Figure 4C) was significantly lower in the WUAP after the third apnea compared with AP (150.90 ± 31.84 ml vs. 177.87 ± 44.50 ml, $p = 0.027$), and in the WUAP (159.71 ± 37.63 ml, $p = 0.005$) than in the CON (209.94 ± 33.84 ml) pre-trial. Spleen volume was also significantly lower (20–37%) from baseline to post-trial in all interventions (CON [$P = 0.013$], WU [$P = 0.001$], AP [$P = 0.001$], and WUAP [$P < 0.001$]).

No significant differences were found across the 4 conditions for BL (Table 2) and Hb (Table 3) when comparing the same time points across the protocol.

Apnea duration showed no differences when comparing the same time point in both interventions. However, in the WUAP, each apnea was longer than the one preceding it (first to second apnea [$P < 0.001$] and second to third apnea [$P = 0.016$]). This was similar in the AP where the first apnea was significantly shorter than the second apnea ($p = 0.014$) and third apnea ($P = 0.014$); nevertheless, the third apnea was not significantly longer in duration than the second ($p = 0.06$) (Figure 5).

Discussion

This is the first study to show that a combination of a warm-up with a series of 3 apneas can improve 400-m freestyle swimming performance against a control.

The finding that a warm-up and apnea combination improves performance against a control is not consistent with previous research (30). Sperlich et al. (30) found no subsequent performance improvements after performing 4 maximal apneas after a warm-up.

It would be expected at a competition that an athlete would perform a warm-up before a race, and although it was not significant in this study, the warm-up improved performance by 0.95% compared with the CON. Yet, with the addition of a series of 3 apneas alongside a warm-up, performance was improved by 1.03% against the CON, yielding a statistically significant outcome. In elite sport, it has been shown that marginal improvements are key to increasing a swimmers chance of achieving an Olympic medal (26). Pyne (26) originally showed that a swimmer should improve their performance by 1% within a competition, then approximately 1% within the year leading up to the Olympics. A further enhancement of 0.4% would substantially increase the medal prospects of the swimmer. When putting this 1.03% improvement in the context of competition results, the improvement would have potentially allowed the second-placed swimmer in the 400-m freestyle at both the World Swimming Championships 2015 and 2016 Rio Summer Olympic Games to win the gold medal.

The stroke parameter results in this study agree with Sperlich et al. (30) by finding that the stroke became more efficient in the WUAP. Initially, in the seventh 50-m repeat, both DPS and SR increased in the WUAP group. Stroke index was improved by 12% ($P = 0.038$) during the last 50-m repeated compared with the CON. This improvement was due to the increased swimming speed, as well as a significantly longer DPS, leading to a more efficient stroke. It has been shown by Laffite et al. (17) that estimated anaerobic contribution increases in the final 100-m of a 400-m freestyle time-trial. Although the lower BL was not statistically significant, this could potentially explain why the post-time-trial BL was 18.96% lower in the WUAP than the WU condition.

When looking at the physiological results, one difference in the results could be caused by the variation of the method, of which splenic volume was calculated (20) in the study by Sperlich et al. (30). This method uses a 2-dimensional measurement to calculate spleen volume, and although this has shown to have a strong positive correlation with actual spleen volume, the newly developed Pilström equation (28) has been specifically developed for the spleens' irregular shape by adding a third dimension into the equation. Results from previous research (9,25) have shown that a typical spleen volume in adults is usually between 200 and 250 ml, which has been supported by analysis of the participants in this study. However, Sperlich et al. (30) found different results with the largest mean spleen volume being 72 ± 10 ml.

Similarly, this study and the study by Sperlich et al. (30) found no change in Hb levels. One proposed explanation for not finding a change in the Hb levels after the splenic contraction could be that suggested by Schagatay et al. (28). They showed that trained free divers have a stronger splenic contraction compared with their untrained counterparts and thereby improve their erythrocyte release. In this study, the participants were not specifically trained in performing maximal apneas. To expose the participants to the feelings of hypercapnia and hypoxia, they had 1 familiarization session until they felt comfortable with the protocol. If they had been provided with a longer period of apnea training as in the study by Engan et al. (7), the physiological response may have been different.

Another potential reason for not seeing an Hb change could be the timing of the splenic contractions occurring and the taking of blood measurements. Studies have shown that a peak can occur up to 15–25 minutes after the spleen has contracted (31). It is believed that this delay is due to the equilibrium between the splenic reservoir and the venous circulatory blood pool; yet, there is no substantive evidence to back this claim (14).

Because of the participants having had no previous experience in apnea training, hyperventilation before the apneas was not used. This could be another reason as to why Hb levels did not change, with studies showing that blood plasma volume is reduced from the previous hyperventilation, which resulted in a higher Hb concentration (29).

Practical Applications

The inclusions of apneas into training should be performed with caution and under the supervision of a trained individual (apnea coach). This is due to their potential hazards such as blackouts that are associated with this method of training, especially with those not experienced in maximal breath-holds. Once competent and the swimmer is confident and comfortable with apneas, they could be integrated into the swimmers' pre-race routine for a 400-m freestyle swim. Once integrated, the current results show that the swimmer could use this method to potentially enhance their 400-m freestyle performance.

This study has shown that a combination of a warm-up with a series of 3 apneas (WUAP) can improve 400-m freestyle swimming performance against a control intervention (CON). The WUAP was shown to be significantly faster than CON and apnea (AP) intervention.

Physiologically, there was no change in the level of Hb; yet, there was a significant reduction in spleen volume in the WU, AP, and WUAP conditions, suggesting the occurrence of a splenic contraction.

The findings of this research support the need for further investigation to find ways of optimizing the use of apneas to enhance performance.

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